

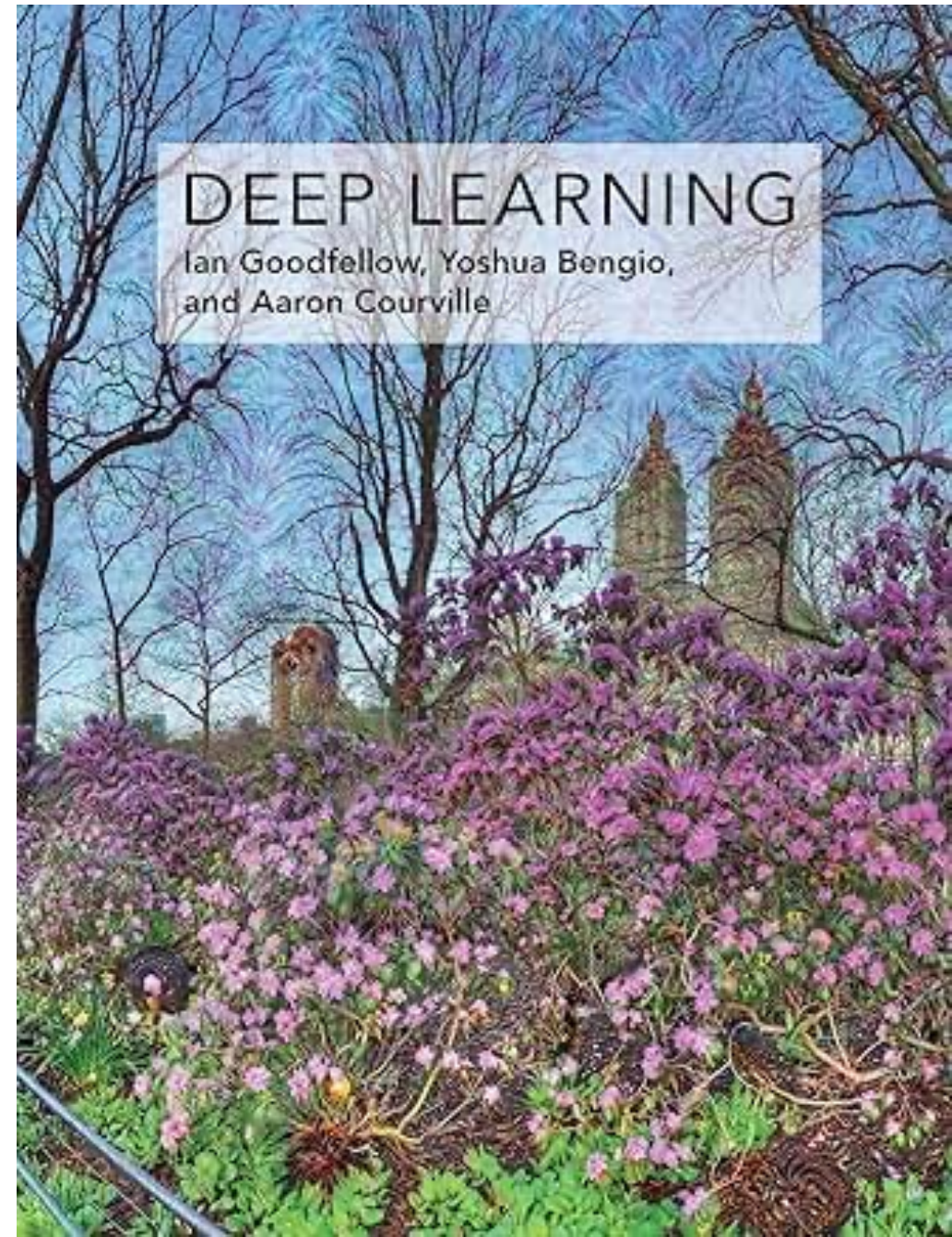


CSCI 6032: Lecture 2 Kinds of ML, Linear Algebra

Instructor: Dr. David A. G. Harrison

Things I want to cover today...

- Types of machine learning
- Start review of linear algebra
- HW 1



Office Hours

207 Weir Hall

Monday	11:00 AM-12:00 PM
Wednesday	1:00-2:00 PM

Google Colab

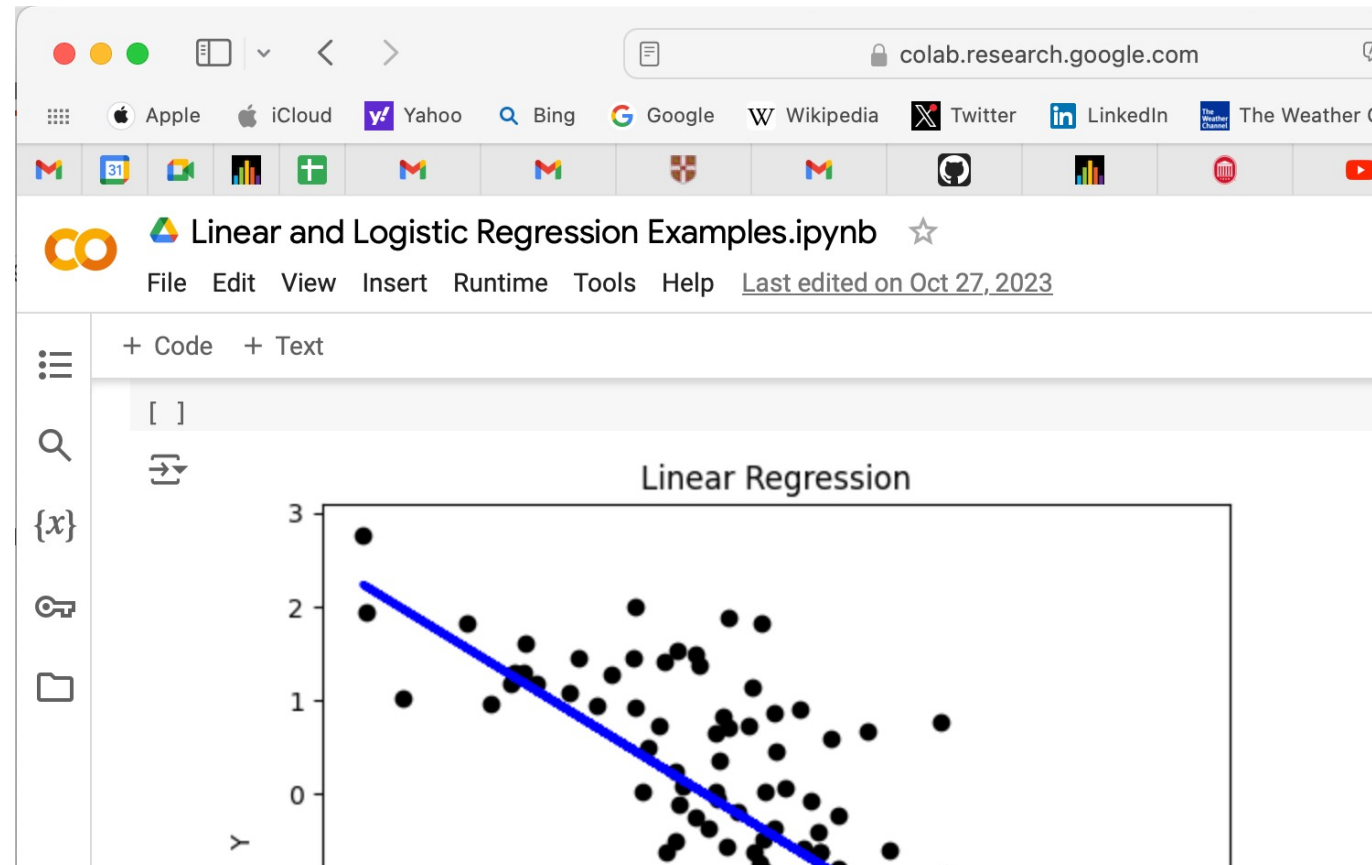
Notebooks and source code examples will also be committed to GitHub.

https://github.com/dosirrah/CSCI6032_26F_MachineLearning

Free.

Sufficient resources for the problems we will tackle.

All homeworks will either be scanned by hand or as a notebook.



Jupyter notebooks

If you like to run things locally. Jupyter notebooks.

In Mac OS, I recommend using Conda

<https://docs.anaconda.com/miniconda/>

Once conda is installed

```
$ conda create -name myenv
```

```
$ conda activate myenv
```

```
$ conda install jupyter
```

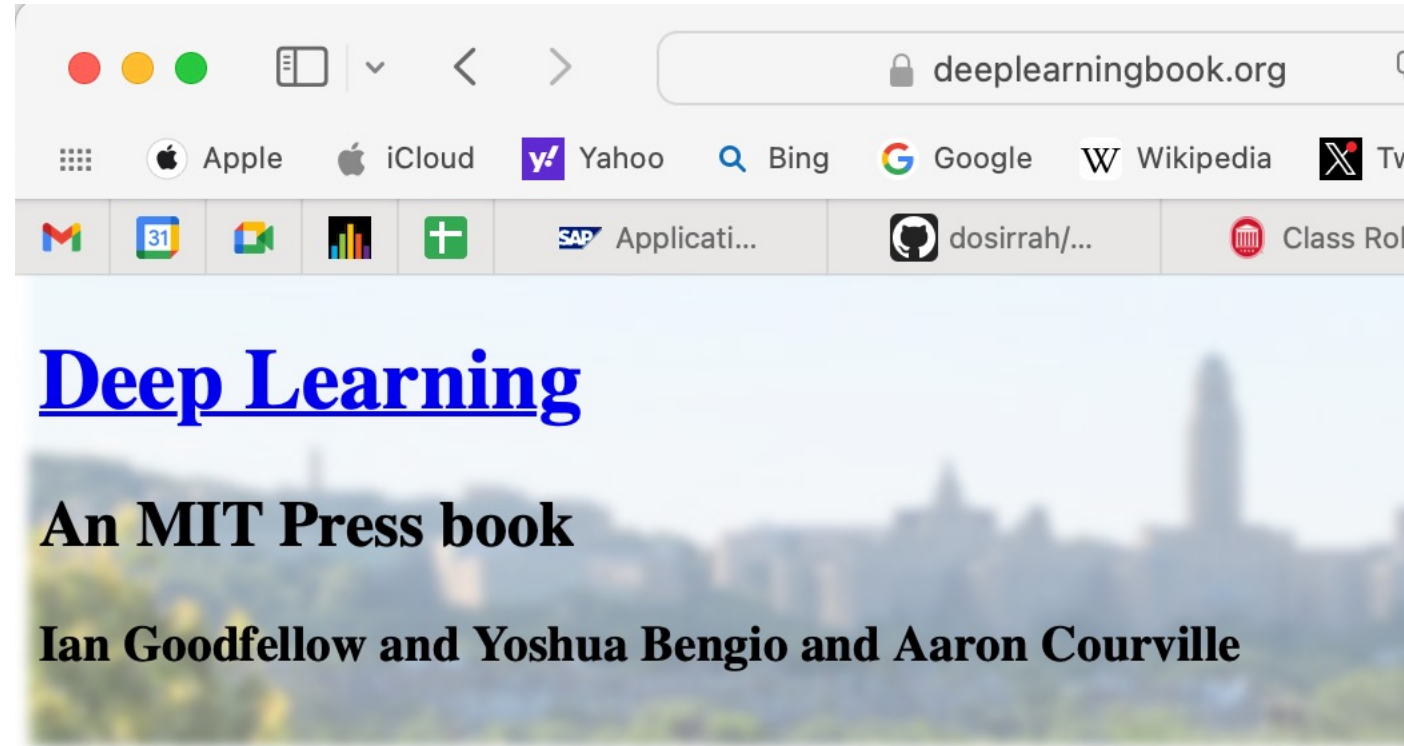
Then to run jupyter notebook.

```
$ jupyter notebook
```


Class Textbook

- ***Available
online for
free***

deeplearningbook.org



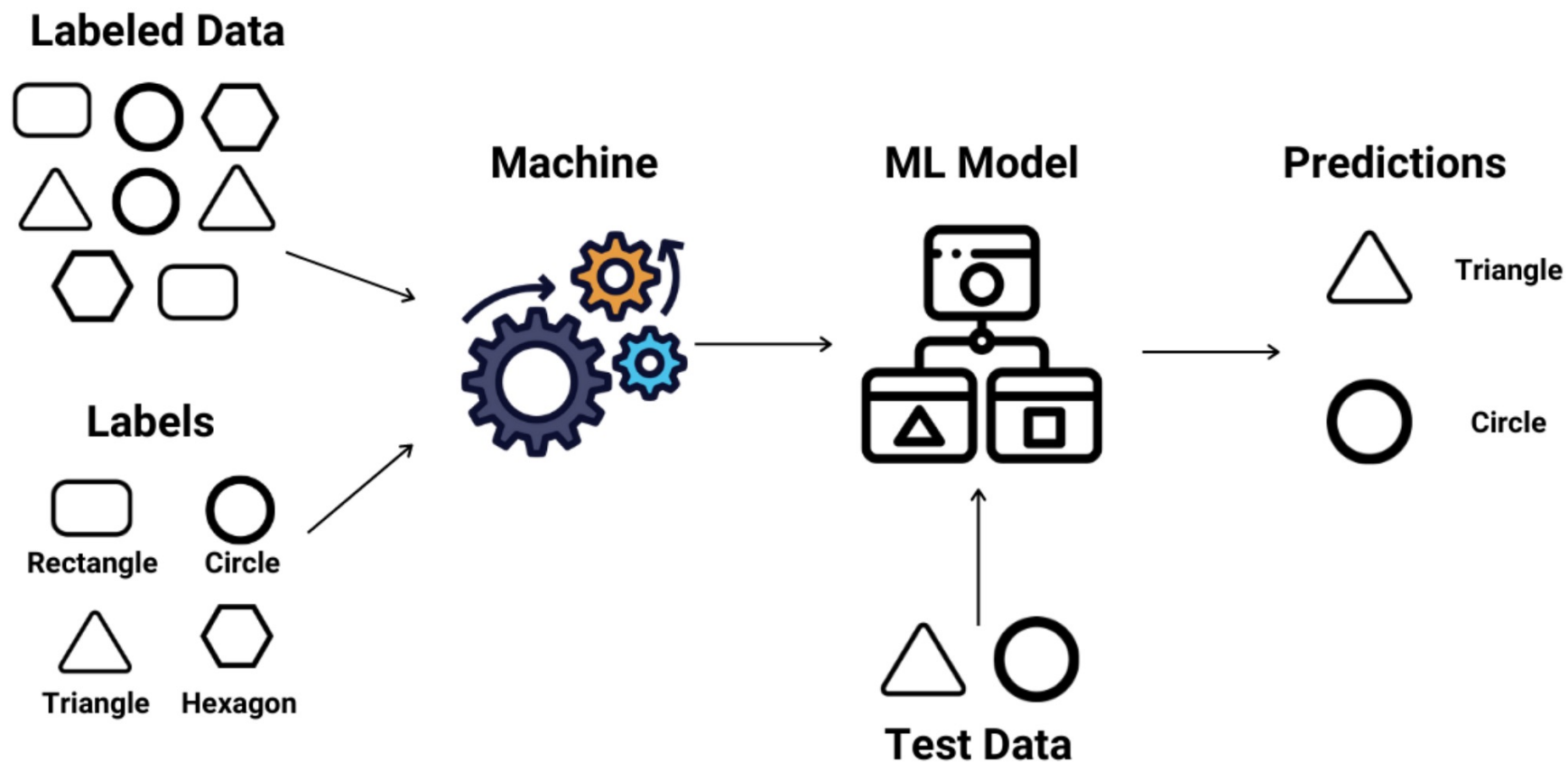
[Exercises](#) [Lectures](#) [External Links](#)

The Deep Learning textbook is a resource intended to help students and practitioners learn about machine learning in general and deep learning in particular. The online version of the book is now completely available for free.

The deep learning textbook can now be ordered on [Amazon](#).

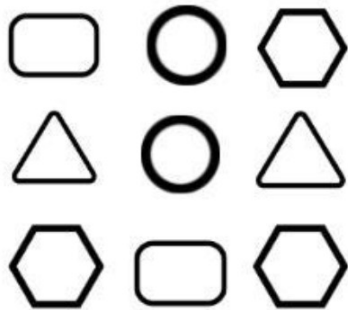
For up to date announcements, join our [mailing list](#).

Supervised Learning



Unsupervised Learning

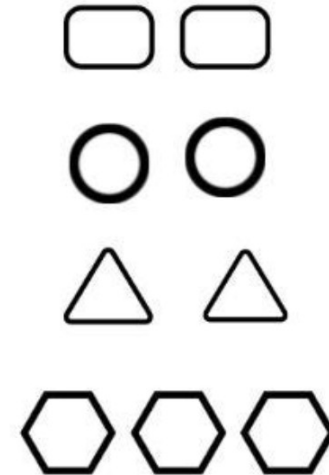
Unlabelled Data



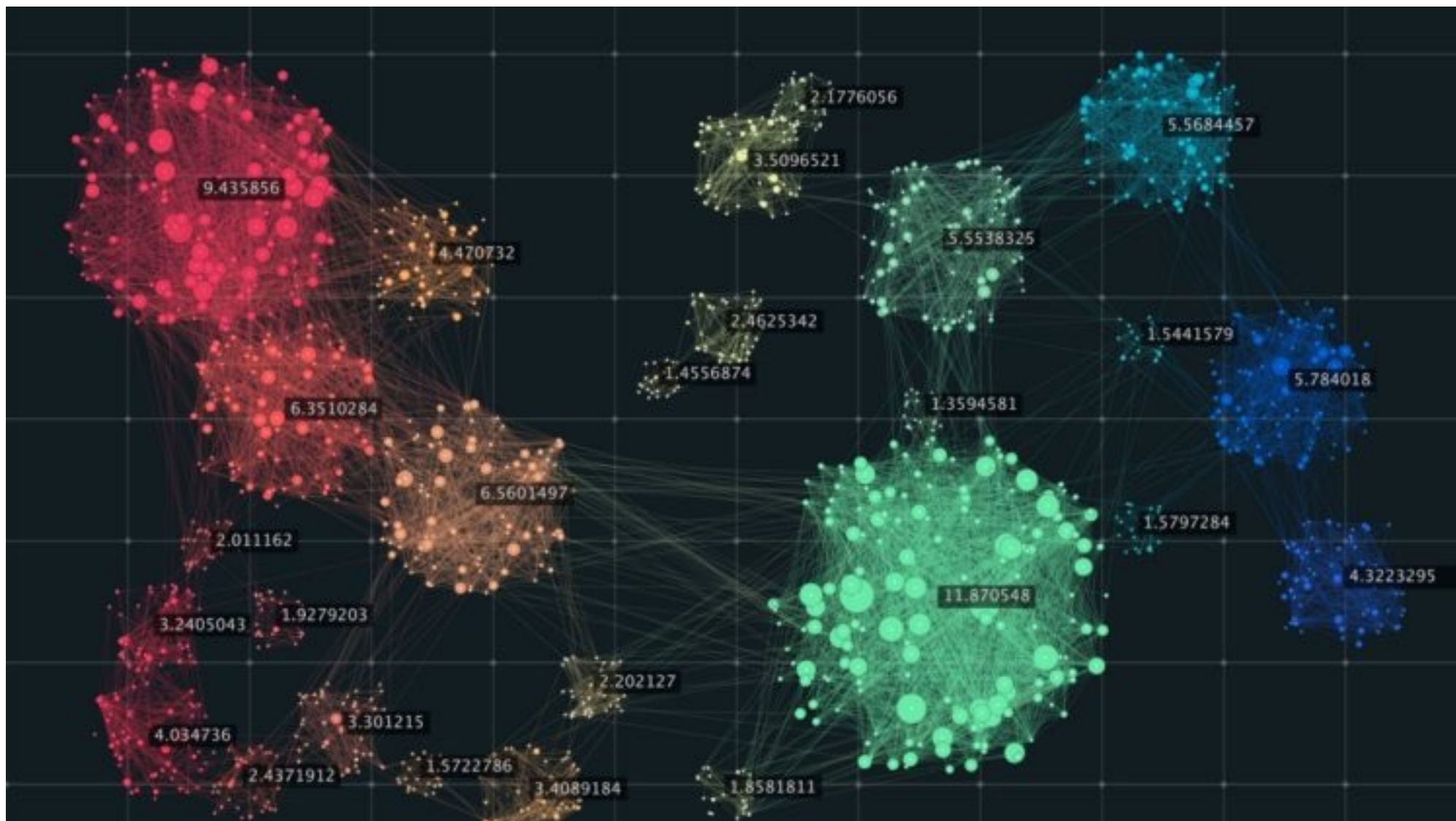
Machine



Results



Unsupervised Learning



Self-supervised Learning

Model learns to predict part of the data from other parts of the data, creating its own supervisory signal without requiring human-provided labels.

1, 2, 3, 4 $\rightarrow$?

2, 3, 5, 7, 11, 13, $\rightarrow$?

Self-supervised Learning

Predict next
move from
prior moves.

NVIDIA Corporation (NVDA)

☆ Follow

+ Add holdings

178.78 **-2.82 (-1.55%)**

As of 9:51:05 AM EDT. Market Open.

 [Time to buy NVDA?](#)

↻ NVDA Q2 2026 Earnings call

Replay the full call for the latest updates and insights.

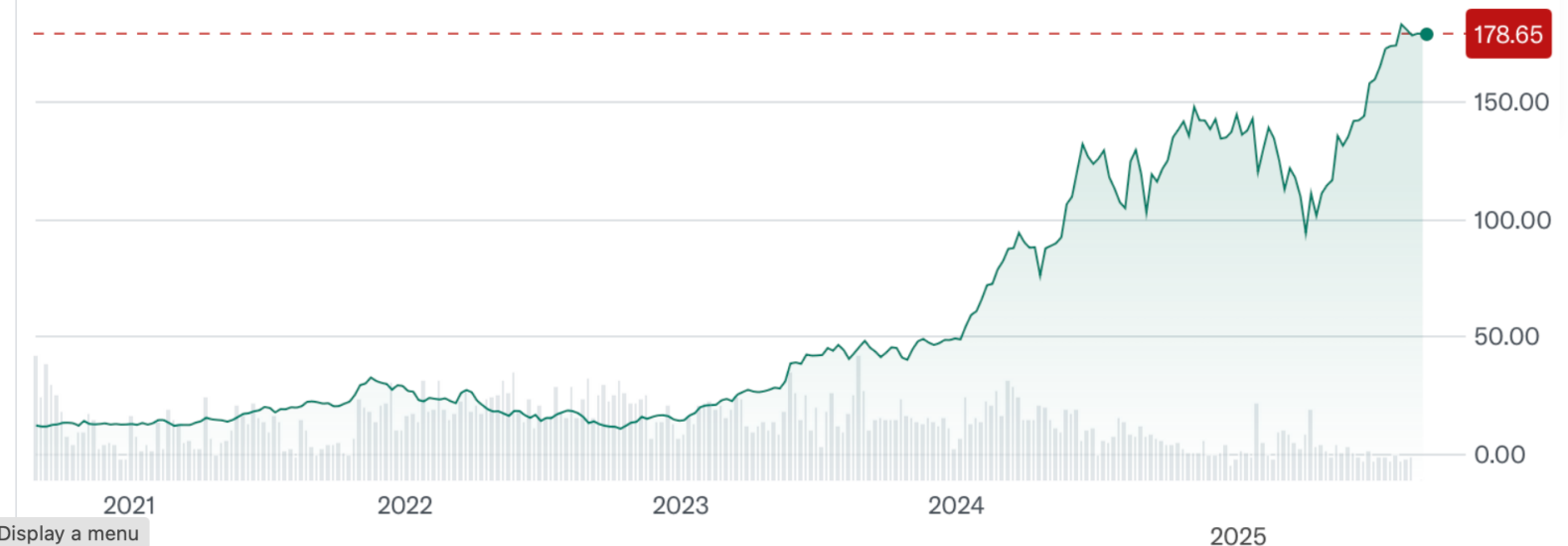
 Read transcript

 Play earnings call

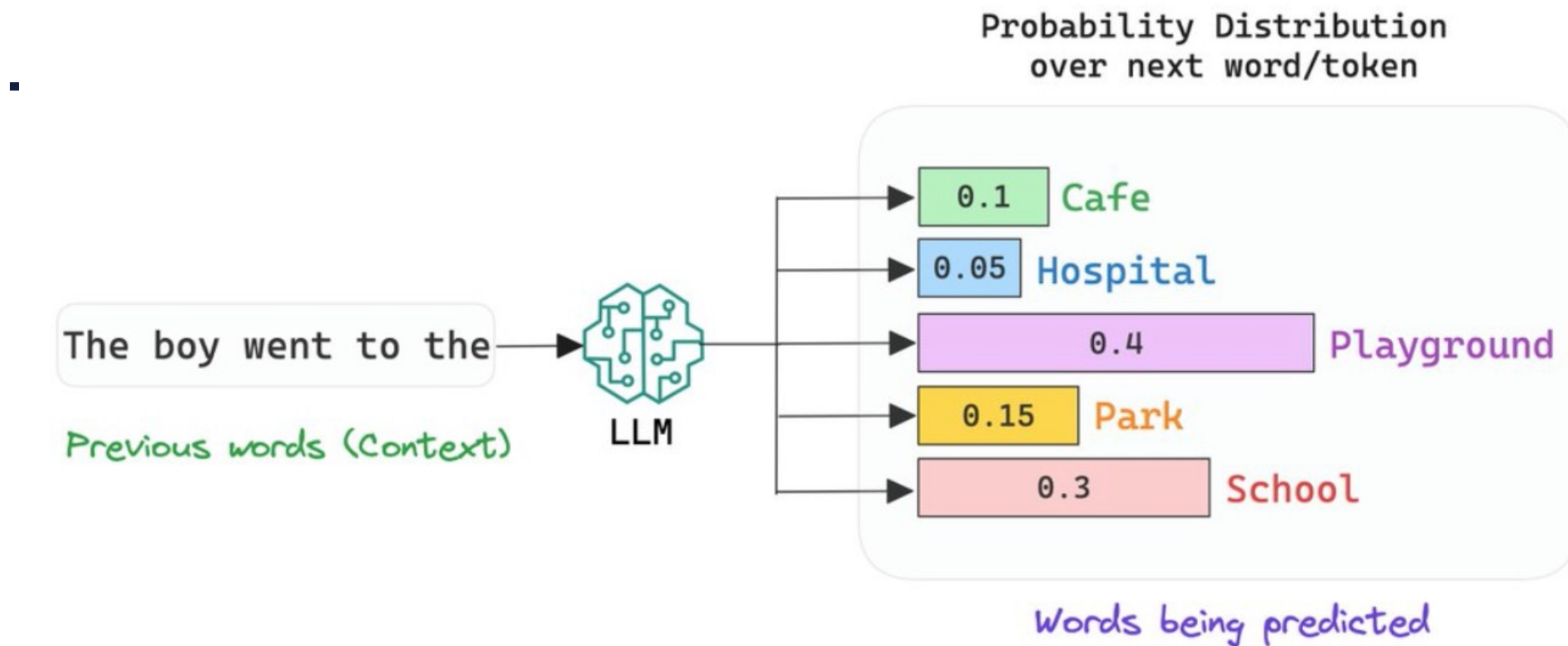
Listen to the full call replay
for the latest updates and
insights

☒ Key Events

▲ Mountain ▼



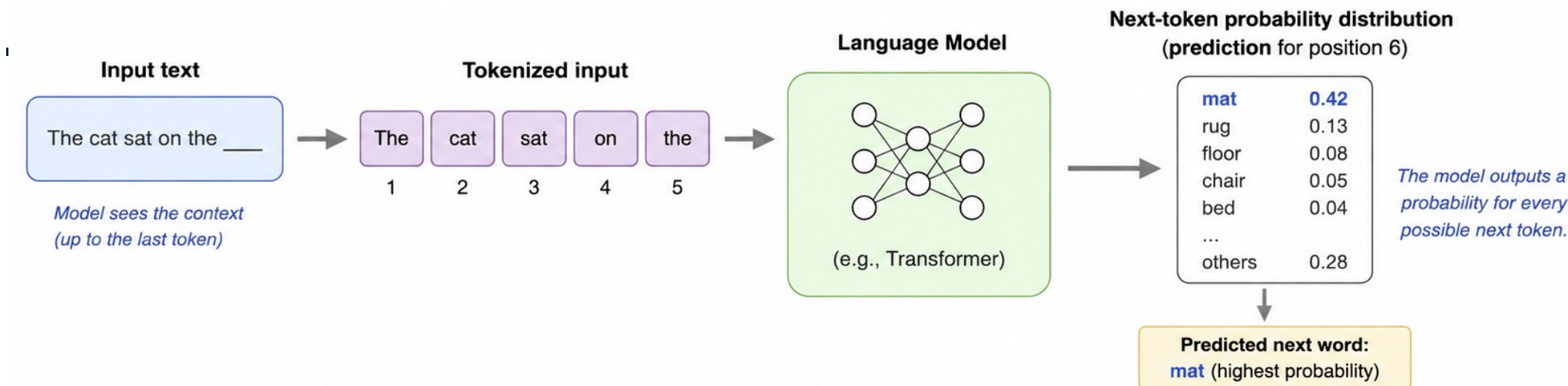
Self-supervised Learning: LLM (1/2)



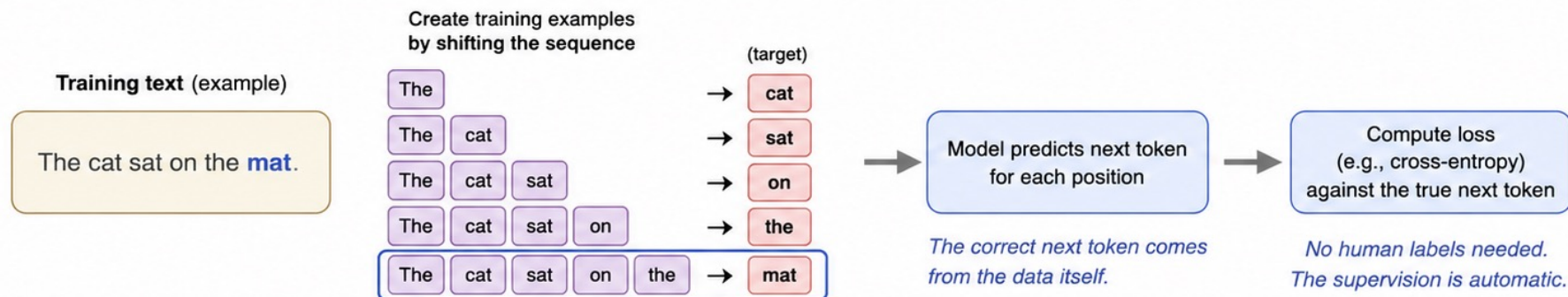
$$P(\text{'Playground'}/\text{'The boy went to'}) = 0.4$$

Self-supervised Learning: LLM (2/2)

Next-Word (Next-Token) Prediction

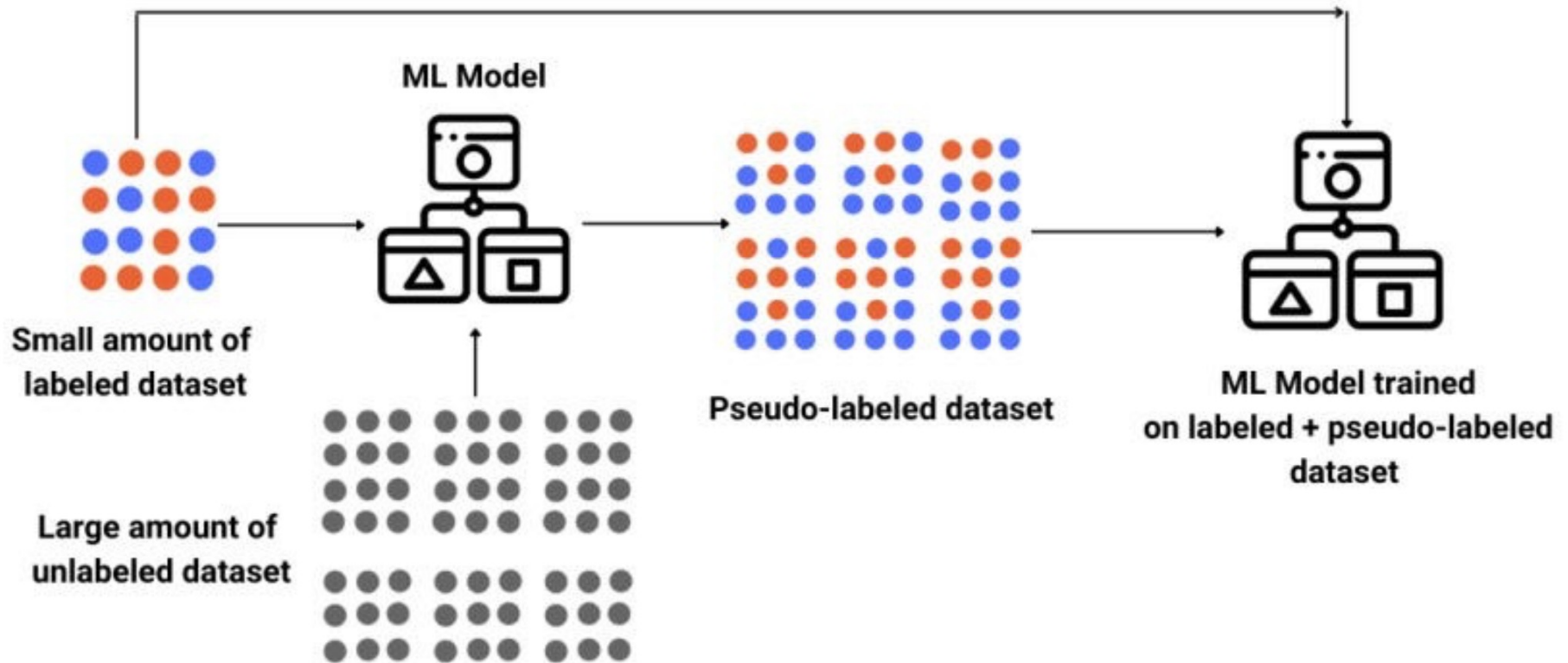


During training (self-supervised)



Summary: The model learns to predict the next word (token) given previous context. The training signal (the "correct" next word) is provided by the text itself → self-supervised learning.

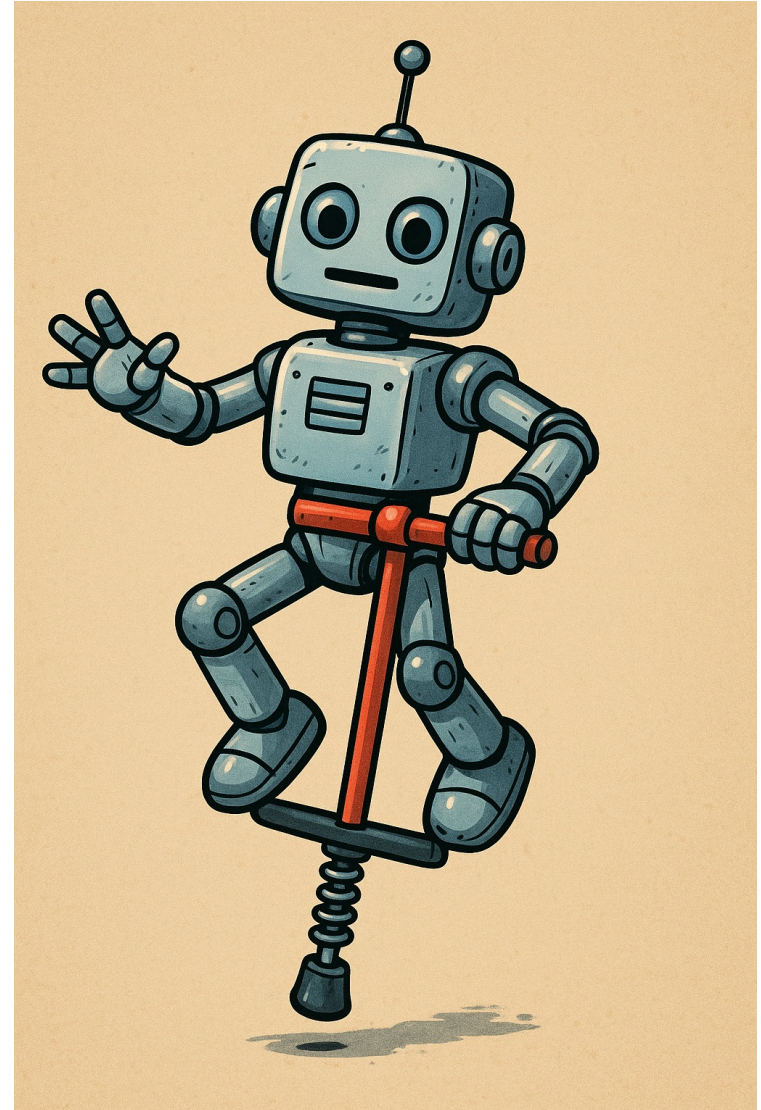
Semi-supervised Learning



Reinforcement Learning

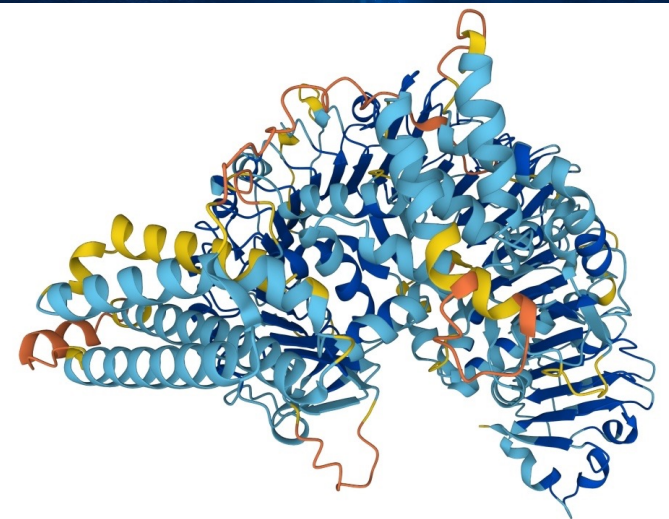
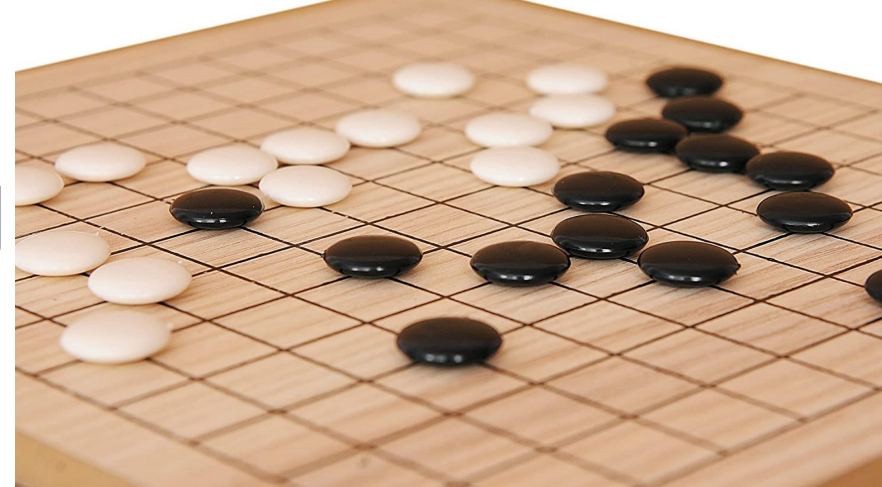
Frame the problem as a game.

A player (a.k.a., agent) makes decisions (i.e., plays the game) and observes the resulting state (e.g., a score).



Reinforcement Learning

- AlphaGo (2016) → beat world champion Lee Sedol at Go.
- AlphaZero (2017) → mastered chess, shogi, and Go from scratch via self-play.
- AlphaFold 1 (2018) → won CASP13 using RL-like
- AlphaStar (2019) → reached Grandmaster level in *StarCraft II*.
- AlphaFold 2 (2020) → solved protein structure prediction. (but no RL!) →



Reinforcement Learning

When trades are large the market starts to look like a game.

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☆ Follow

+ Add holdings

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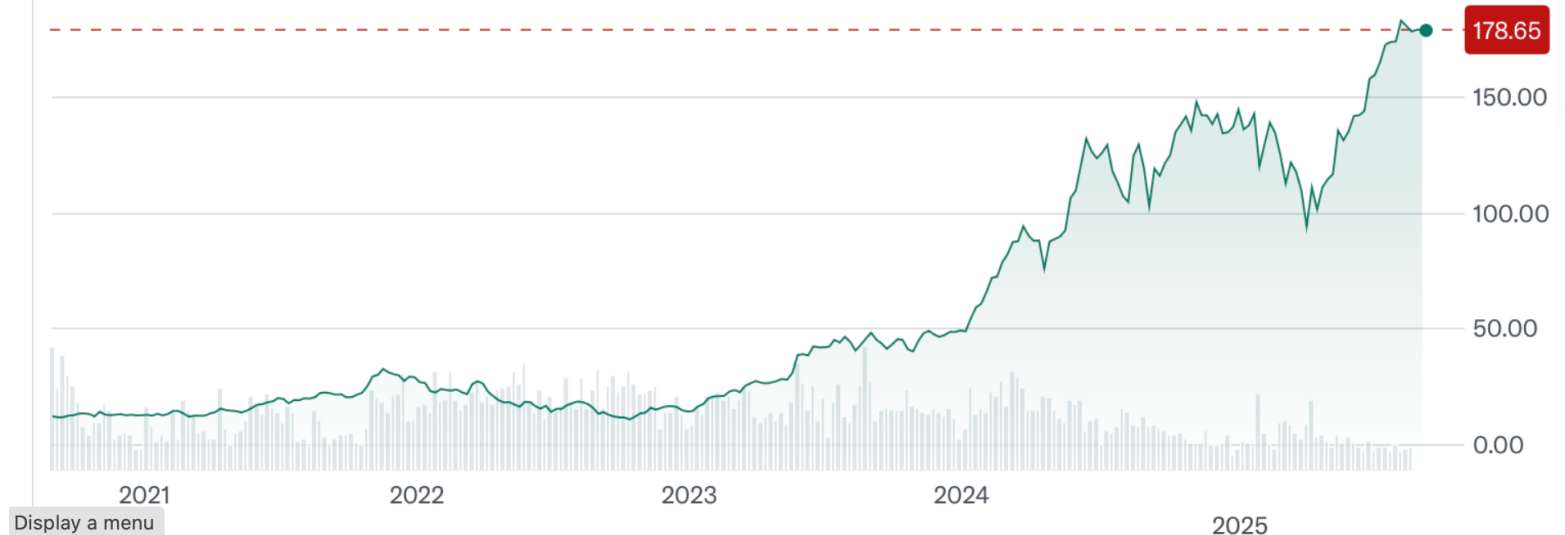
 Play earnings call

Listen to the full call replay for the latest updates and insights



☒ Key Events

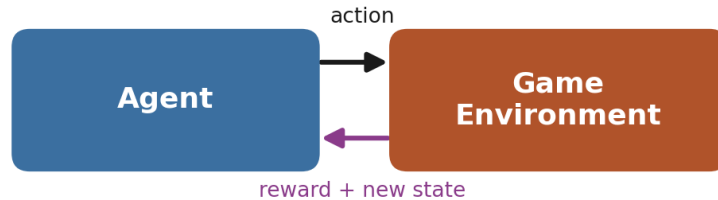
▲ Mountain ▼



Display a menu

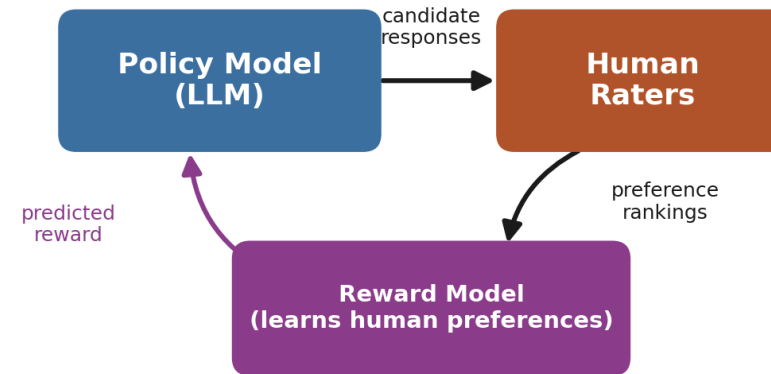
RLHF (Reinforcement Learning from Human Feedback)

Reinforcement Learning in Games (e.g., AlphaGo)



Reward = game's built-in score (win/loss, points) — objective, fixed by the rules of the game

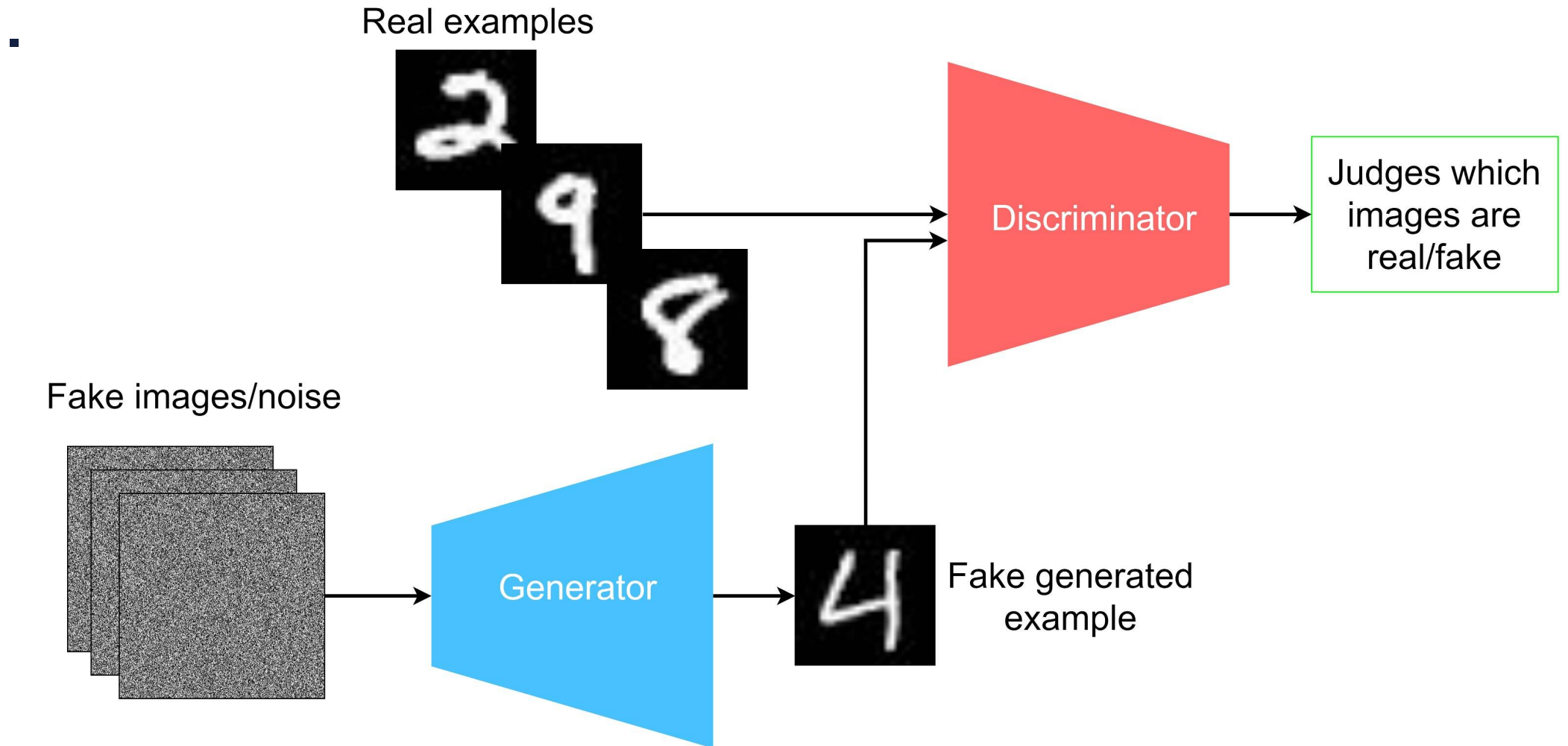
RLHF (e.g., training a chat model)



Reward is learned from human judgments (subjective, no built-in scoring function like a game has)

Key difference: in games, the environment hands back an objective reward. In RLHF, a learned reward model stands in for human judgment — there's no rulebook that scores "helpfulness."

Generative Adversarial Networks



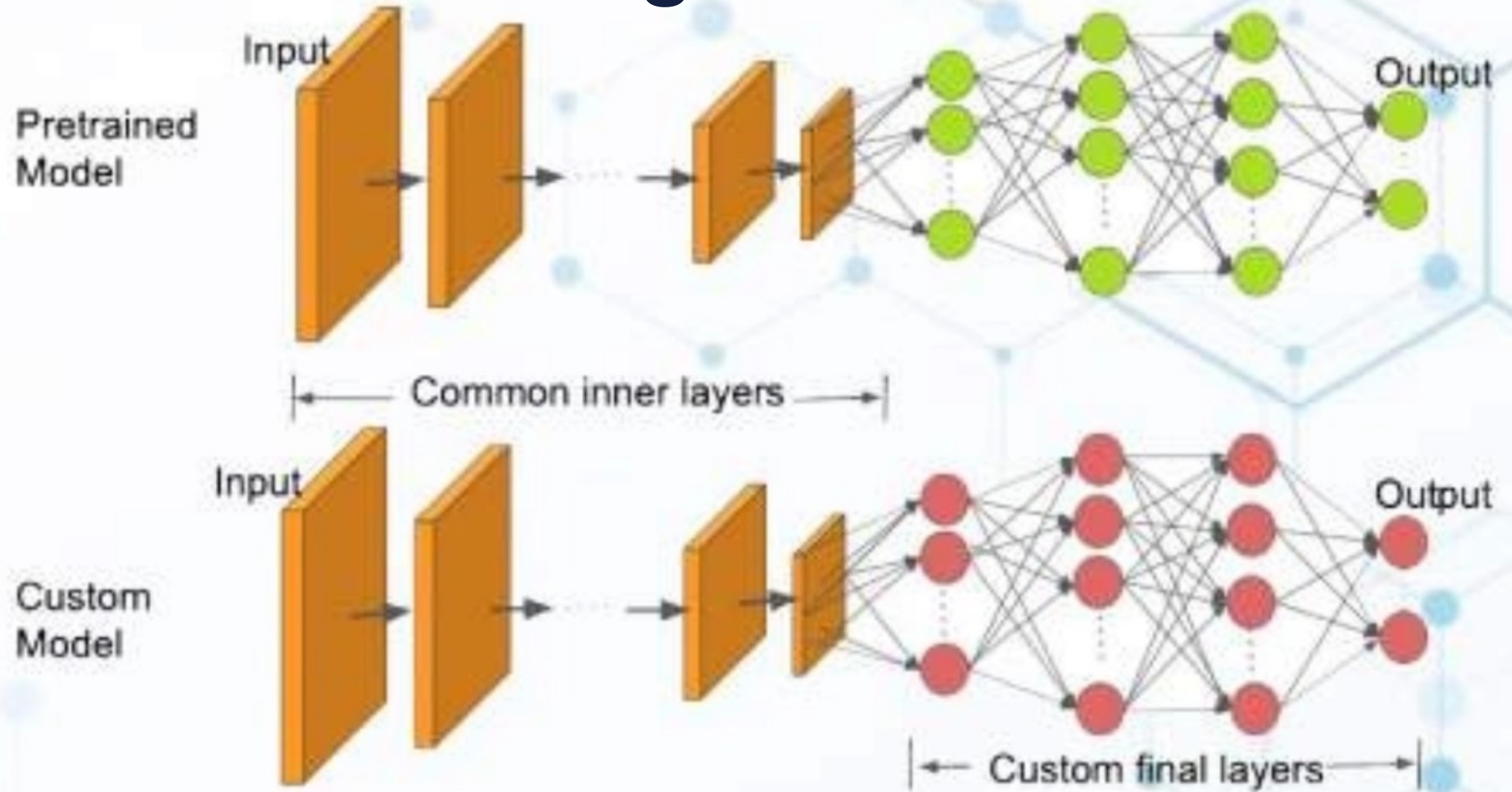
Generative Adversarial Networks

Ex: Generative art from an
encoder trained using a GAN.

“Draw with hyperrealism a cat
with a crocodile head with a
rose in its mouth dancing”



Transfer Learning



Ensemble Methods

Combine predictions from many models to do better than any single model alone.

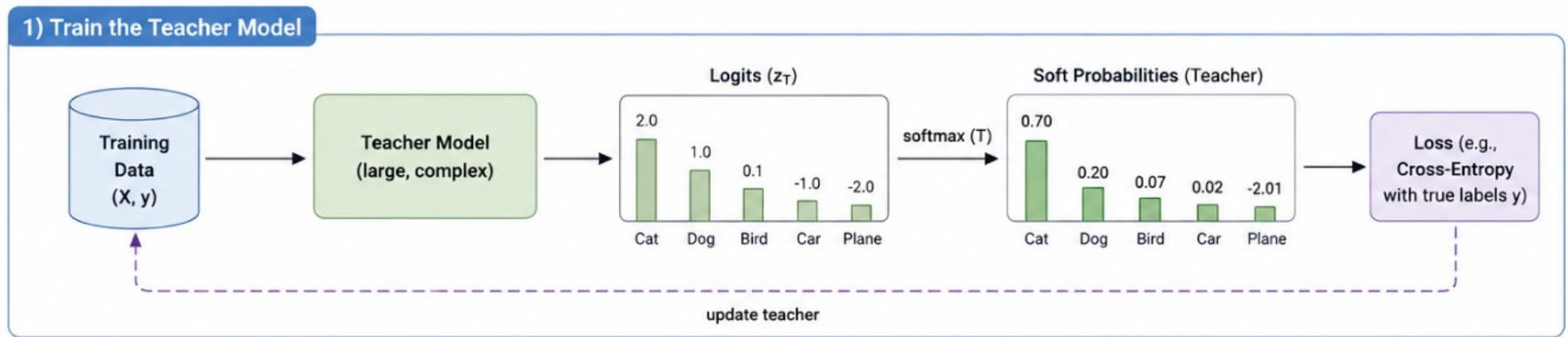
Bagging (Bootstrap Aggregating): train models independently on random subsets of the data, then average/vote their predictions. Ex: Random Forest (later in the course).

Boosting: train models sequentially, where each new model focuses on correcting the previous models' mistakes. Ex: AdaBoost, Gradient Boosting / XGBoost.

Distillation (1/2)

Train a large model as teacher

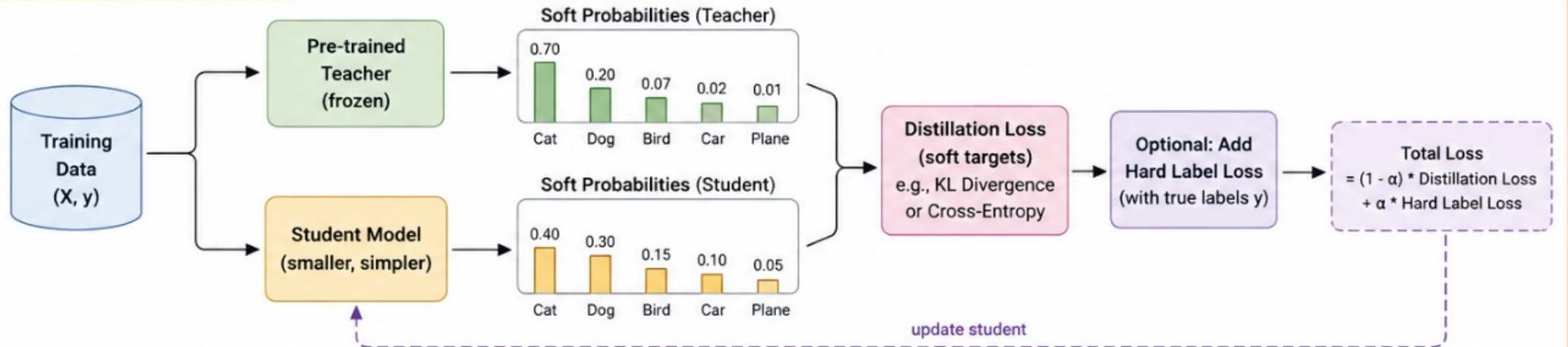
Ex: frontier model (GPT, Gemini, Grok, Llama, DeepSeek, Qwen, Mistral, Kimi, ..)



Distillation (2/2)

Teacher teaches smaller model

2) Distill to the Student Model



Why USE PYTHON FOR Machine learning?

Python is easy to program.

Python has beautiful, terse syntax.

Python is well supported by tools.

The big stuff is usually not done in Python, so slow doesn't matter (usually):

- **PyTorch**
- **TensorFlow**

The medium-sized stuff and postprocessing is usually not done in Python either

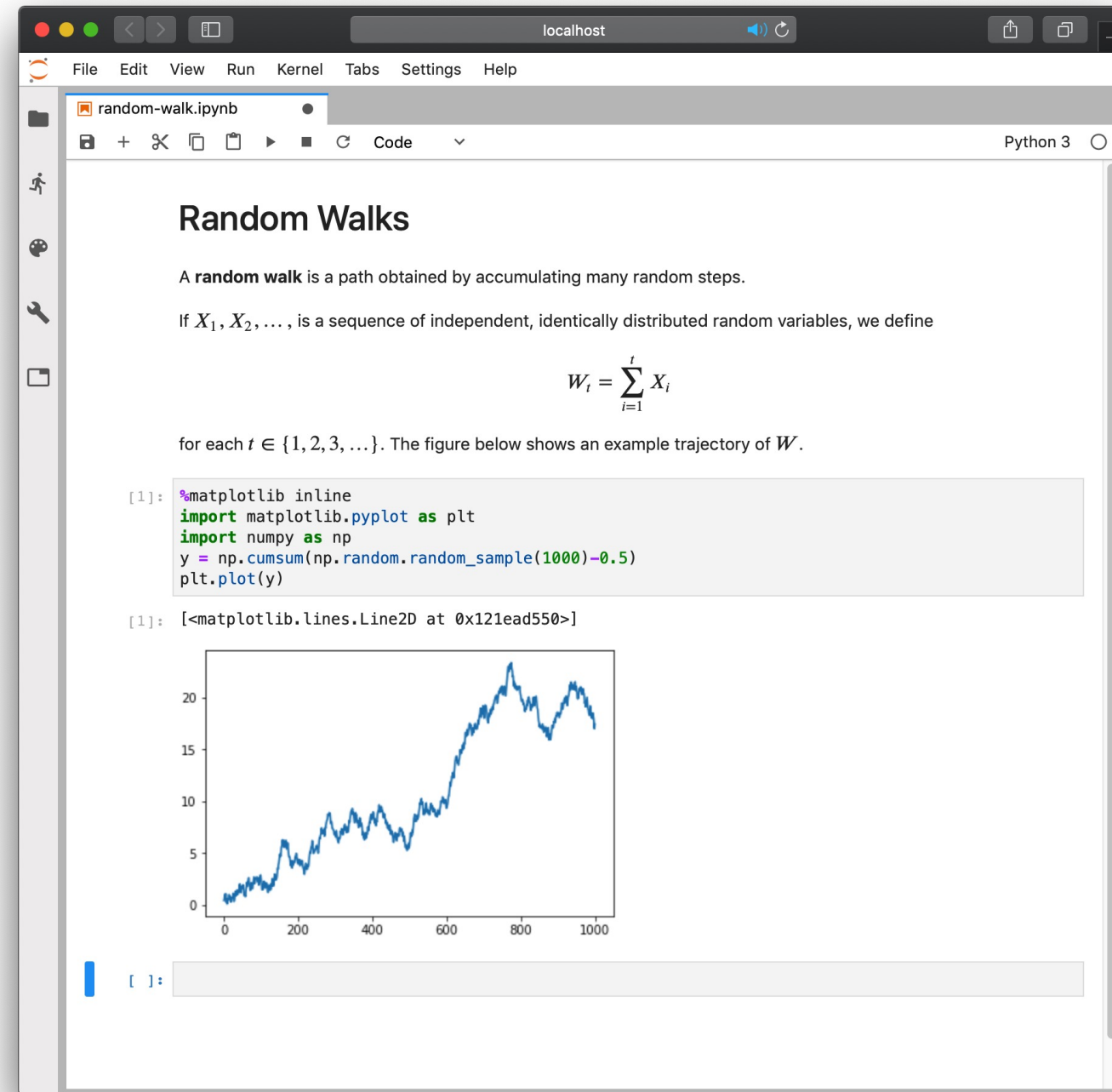
- **NumPy**
- **DataFrames**
- **SciPy**

Why use jupyter?

All of the benefits of Python.

The ability to mix visualization, description (markdown), math (latex), and code in the same environment: a notebook.

Ability to run the front end locally or remotely.

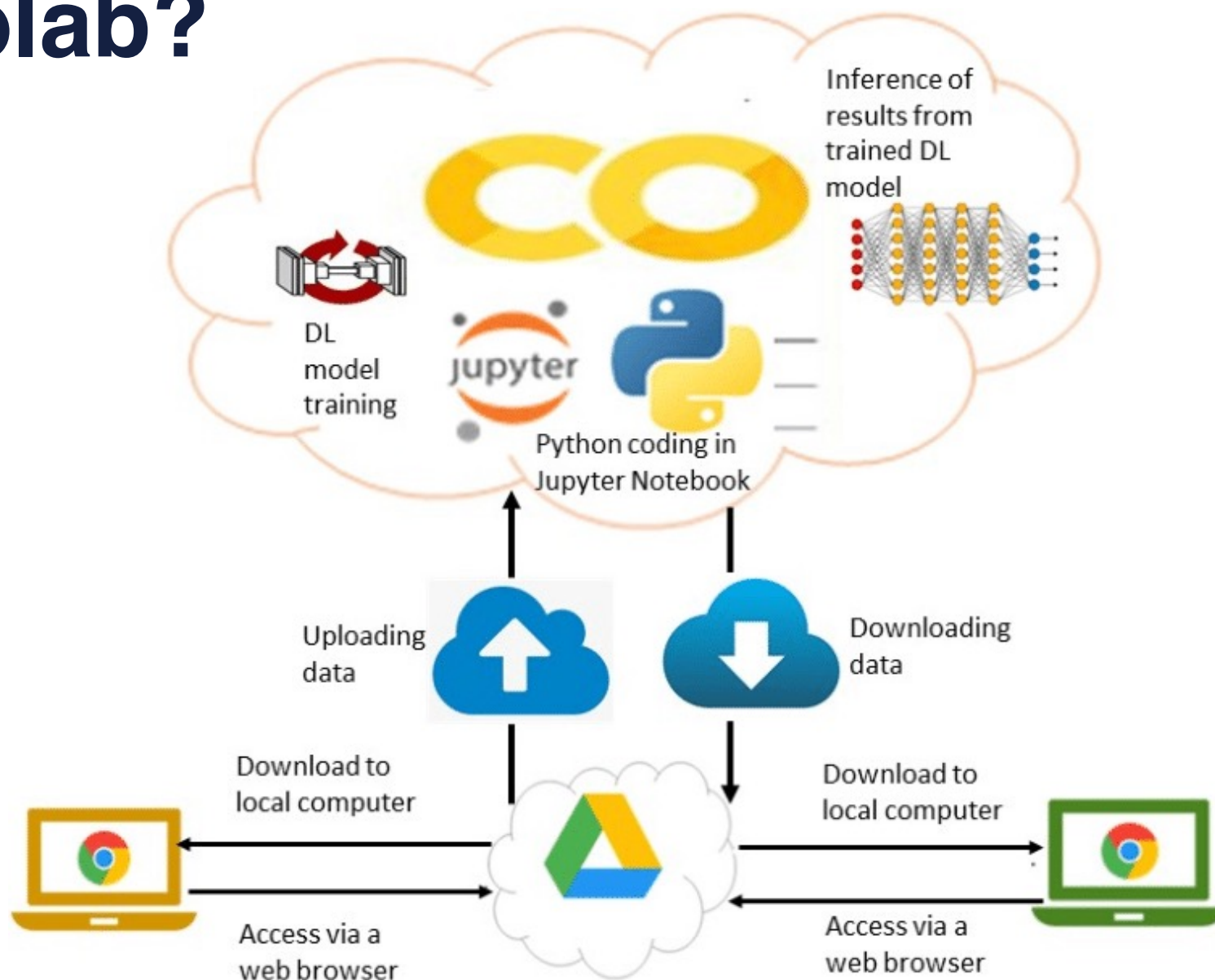


Why use Colab?

All of the benefits of jupyter and python but run remotely.

Can collaborate by sharing the notebook.

Ability to train and validate machine learning algorithms across a cluster (e.g., using PyTorch or TensorFlow) while displaying the training and validation in a notebook.

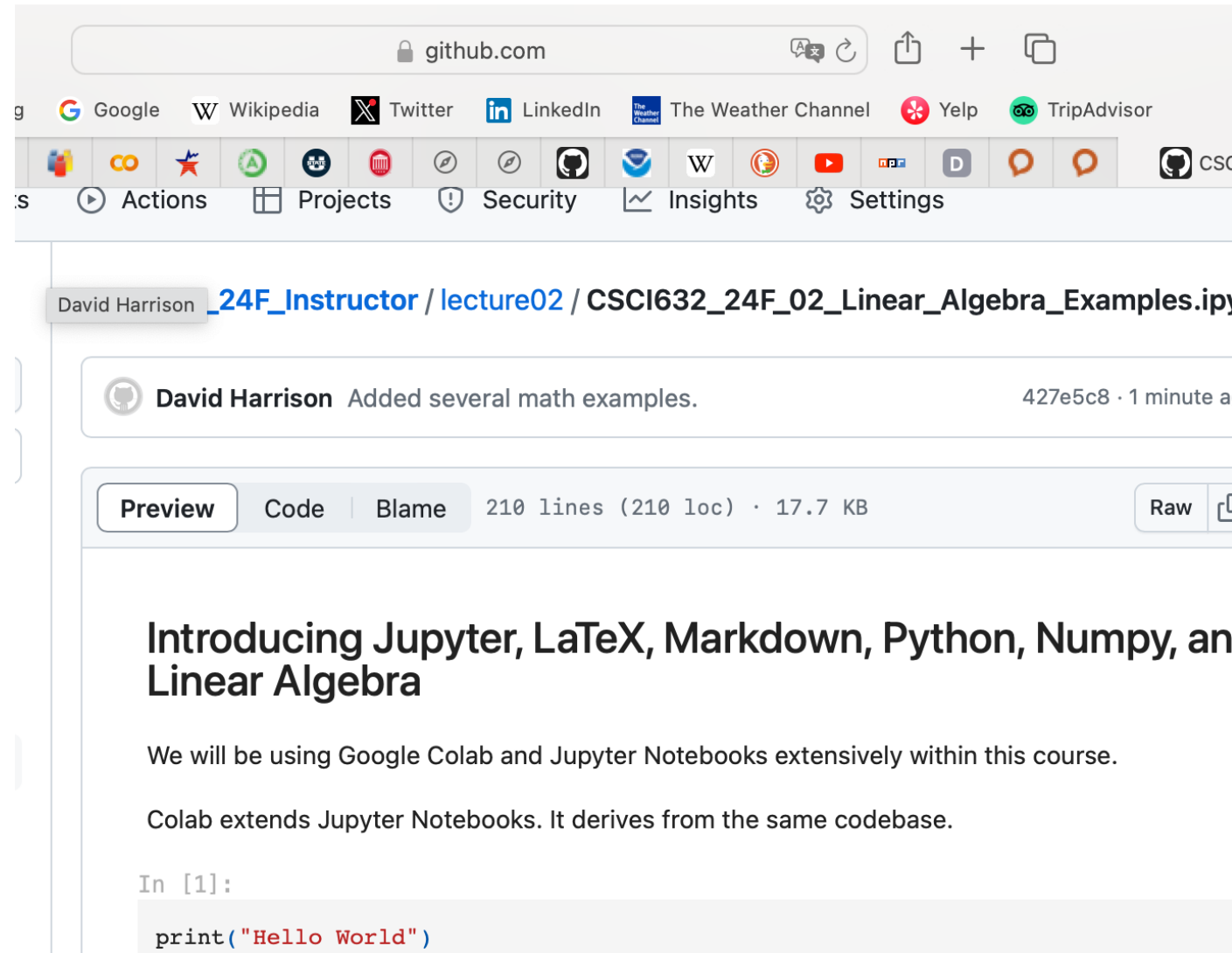


Jupyter and Latex

See Jupyter notebook
Similar can be done
in Google Colab

You can open this
notebook in GitHub
WITHOUT installing
jupyter on your local
system.

https://github.com/dosirrah/CSCI6032_26F_Instructor/blob/main/lecture02/CSCI6032_26F_02_Linear_Algebra_Examples.ipynb



The screenshot shows a web browser displaying the GitHub interface. The address bar shows 'github.com'. The page header includes navigation links like 'Actions', 'Projects', 'Security', 'Insights', and 'Settings'. The main content area shows a commit by 'David Harrison' titled '_24F_Instructor / lecture02 / CSCI632_24F_02_Linear_Algebra_Examples.ipynb'. Below the commit, there are tabs for 'Preview', 'Code', and 'Blame'. The 'Preview' tab is selected, showing the content of the Jupyter notebook. The notebook title is 'Introducing Jupyter, LaTeX, Markdown, Python, Numpy, and Linear Algebra'. The text in the notebook says: 'We will be using Google Colab and Jupyter Notebooks extensively within this course. Colab extends Jupyter Notebooks. It derives from the same codebase.' Below this, there is a code cell labeled 'In [1]:' containing the code 'print("Hello World")'.

github.com

Google Wikipedia Twitter LinkedIn The Weather Channel Yelp TripAdvisor

Actions Projects Security Insights Settings

David Harrison _24F_Instructor / lecture02 / CSCI632_24F_02_Linear_Algebra_Examples.ipynb

David Harrison Added several math examples. 427e5c8 · 1 minute ago

Preview Code Blame 210 lines (210 loc) · 17.7 KB Raw

Introducing Jupyter, LaTeX, Markdown, Python, Numpy, and Linear Algebra

We will be using Google Colab and Jupyter Notebooks extensively within this course.

Colab extends Jupyter Notebooks. It derives from the same codebase.

In [1]:

```
print("Hello World")
```


Homework 0 (ungraded, handed out last time)

1. Read Chapter 1 and start reviewing linear algebra in chapter 2.
2. Confirm you can login to blackboard.
3. Setup an account with github
<https://github.com/>
4. Checkout the class repository locally
https://github.com/dosirrah/CSCI6032_26F_MachineLearning
5. Create an account with google colab.
6. (optional) Setup a local environment using conda and jupyter.

A vertical banner on the left side of the slide with the word "HOLLYWOOD" written vertically in white capital letters on a red and blue background.

HW1

On github already...

Due next Thursday

Submit as notebook via blackboard.

Notation

We will use the notation from Goodfellow et al, *Deep Learning*.

To the right is just the first page of notation from the book.

<https://www.deeplearningbook.org/contents/notation.html>

Numbers and Arrays

a	A scalar (integer or real)
$\mathbf{a}$	A vector
$\mathbf{A}$	A matrix
$\mathbf{A}$	A tensor
$\mathbf{I}_n$	Identity matrix with n rows and n columns
$\mathbf{I}$	Identity matrix with dimensionality implied by context
$\mathbf{e}^{(i)}$	Standard basis vector $[0, \dots, 0, 1, 0, \dots, 0]$ with a 1 at position i
$\text{diag}(\mathbf{a})$	A square, diagonal matrix with diagonal entries given by $\mathbf{a}$
a	A scalar random variable
$\mathbf{a}$	A vector-valued random variable
$\mathbf{A}$	A matrix-valued random variable